HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Python** scripts for data processing and automation, handling approximately **1.5 TB** of raw data weekly for analytics workflows.
- Designed and implemented **REST APIs** using **Java Spring Boot** for seamless integration between enterprise systems, supporting over **500,000** daily transactions.
- Contributed to the enhancement of a **payments operations dashboard** by implementing real-time transaction monitoring alerts, reducing critical incident response time by **30%**.
- Performed analytical tasks involving **time-series data** and mathematical models for energy technology optimisation, contributing to improved system performance.
- Wrote, tested, and debugged **Python code** for analytics, data processing, and automation, supporting research and development software prototypes.
- Collaborated with senior staff and customers on software module design and implementation, managing tasks with minimal guidance.
- Maintained clear, concise technical documentation for software modules and data-processing components, ensuring knowledge transfer for cross-functional teams.

SKILLS

Programming Languages – Python, Java, SQL, R, Linux Shell Scripting

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, FastAPI, Flask

Databases & Data Paradigms – PostgreSQL, MySQL, Oracle, NoSQL, Key-Value Stores, Time-Series Databases, Relational Data Modelling

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Tools & Technologies – Splunk, Dynatrace, Jira, Confluence, Power BI, Tableau, RStudio

Concepts – Time-series data, Analytics workflows, Statistical methods, Algorithms, Data structures, API development, Data processing, Optimisation workflows, Containerisation

PROJECTS

Energy Data Analytics Pipeline (HCL Technologies)

- Engineered a **Python**-based data pipeline to ingest and process large volumes of **time-series data** from energy sensors, supporting analytics and optimisation workflows.
- Implemented data validation and cleaning routines using **pandas** and **NumPy**, ensuring data quality for downstream mathematical modelling and analysis.
- Developed **SQL queries** to extract and transform data from relational databases, integrating with **AWS S3** for data storage and retrieval.

System Performance Optimisation Software (HCL Technologies)

- Contributed to the development of software modules focused on system performance optimisation within an enterprise environment.
- Assisted in troubleshooting and debugging issues across software layers, improving system reliability and reducing processing bottlenecks.
- Wrote and maintained technical documentation for the optimisation software, detailing algorithms and implementation strategies for internal teams.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)